

Common Denominators Practice

Answer Key

Grade 4–5

Quick Answers

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|--|---------------------------|
| 1. numerator for $1/2 = ?/6 = 3$, numerator for | 6. $\frac{11}{12}$ |
| $1/3 = ?/6 = 2$ | 7. 40 |
| 2. 12 | 8. $\frac{9}{20}$ |
| 3. $\frac{2}{3}$ | 9. $\frac{13}{24}$ m |
| 4. numerator for $3/4 = ?/20 = 15$, numerator for | 10. $\frac{5}{24}$ kg |
| $2/5 = ?/20 = 8$ | 11. $1 \cdot \frac{1}{3}$ |
| 5. $1 \cdot \frac{1}{2}$ | 12. $\frac{11}{12}$ |
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Curriculum

Level: Grade 4–5

5.NF.A.1 – problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12

Difficulty 1–5 of 5 across 14 tagged checks.

1. Fill in the missing numerators: $\frac{1}{2} = \frac{\square}{6}$ and $\frac{1}{3} = \frac{\square}{6}$.

Solution: Multiply the top and bottom of each fraction by the same number so the denominator becomes 6. For $\frac{1}{2}$: multiply by 3 over 3, so $\frac{1}{2} = \frac{1 \times 3}{2 \times 3}$, and the numerator is 3

For $\frac{1}{3}$: multiply by 2 over 2, so $\frac{1}{3} = \frac{1 \times 2}{3 \times 2}$, and the numerator is 2

2. What is the smallest common denominator you can use for $\frac{1}{4}$ and $\frac{5}{12}$?

Solution: List multiples of 4: 4, 8, 12, ... The number 12 is already a multiple of 4, because $3 \times 4 = 12$. So twelfths work for both fractions, and nothing smaller does, since the denominator must be a multiple of 12. Smallest common denominator: 12

3. Add: $\frac{1}{2} + \frac{1}{6}$

Solution: The smallest common denominator of 2 and 6 is 6. Rewrite: $\frac{1}{2} = \frac{3}{6}$. Then $\frac{3}{6} + \frac{1}{6} = \frac{4}{6}$, which simplifies by dividing top and bottom by 2. Final answer: $\frac{2}{3}$

4. Rewrite both fractions with denominator 20: $\frac{3}{4}$ and $\frac{2}{5}$.

Solution: For $\frac{3}{4}$: since $4 \times 5 = 20$, multiply top and bottom by 5: $\frac{3 \times 5}{4 \times 5}$, numerator 15

For $\frac{2}{5}$: since $5 \times 4 = 20$, multiply top and bottom by 4: $\frac{2 \times 4}{5 \times 4}$, numerator 8

5. Subtract: $\frac{5}{6} - \frac{1}{3}$

Solution: The smallest common denominator of 6 and 3 is 6. Rewrite: $\frac{1}{3} = \frac{2}{6}$. Then $\frac{5}{6} - \frac{2}{6} = \frac{3}{6}$, which simplifies to $\frac{1}{2}$

6. Add: $\frac{2}{3} + \frac{1}{4}$

Solution: The denominators 3 and 4 share no common factor, so the smallest common denominator is $3 \times 4 = 12$. Rewrite: $\frac{2}{3} = \frac{8}{12}$ and $\frac{1}{4} = \frac{3}{12}$. Then $\frac{8}{12} + \frac{3}{12} = \boxed{\frac{11}{12}}$

7. Which of 18, 40, or 80 is the smallest common denominator that works for $\frac{5}{8}$ and $\frac{3}{10}$?

Solution: A common denominator must be a multiple of 8 *and* of 10. Check 18: not a multiple of 8, so it fails. Check 40: $8 \times 5 = 40$ and $10 \times 4 = 40$, so 40 works. Check 80: also works, but it is larger. Smallest working choice: 40

8. Subtract: $\frac{7}{10} - \frac{1}{4}$

Solution: The smallest common denominator of 10 and 4 is 20. Rewrite: $\frac{7}{10} = \frac{14}{20}$ and $\frac{1}{4} = \frac{5}{20}$. Then $\frac{14}{20} - \frac{5}{20} = \boxed{\frac{9}{20}}$

9. Ana ties a ribbon $\frac{3}{8}$ m long to a ribbon $\frac{1}{6}$ m long. How long are the two ribbons together?

Solution: We need $\frac{3}{8} + \frac{1}{6}$. The smallest common denominator of 8 and 6 is 24. Rewrite: $\frac{3}{8} = \frac{9}{24}$ and $\frac{1}{6} = \frac{4}{24}$. Then $\frac{9}{24} + \frac{4}{24} = \frac{13}{24}$. Total length: $\frac{13}{24}$ m

10. A bag holds $\frac{7}{8}$ kg of flour. Leo uses $\frac{2}{3}$ kg. How much flour is left?

Solution: We need $\frac{7}{8} - \frac{2}{3}$. The smallest common denominator of 8 and 3 is 24. Rewrite: $\frac{7}{8} = \frac{21}{24}$ and $\frac{2}{3} = \frac{16}{24}$. Then $\frac{21}{24} - \frac{16}{24} = \frac{5}{24}$. Flour left: $\frac{5}{24}$ kg

11. Find the missing fraction: $\frac{5}{12} + \square = \frac{3}{4}$

Solution: Rewrite $\frac{3}{4}$ in twelfths: $\frac{3}{4} = \frac{9}{12}$. The missing fraction is $\frac{9}{12} - \frac{5}{12} = \frac{4}{12}$, which simplifies to $\frac{1}{3}$

12. Challenge. Add all three: $\frac{1}{2} + \frac{1}{3} + \frac{1}{12}$

Solution: One common denominator works for all three fractions: 12 is a multiple of 2, of 3, and of 12. Rewrite: $\frac{1}{2} = \frac{6}{12}$, $\frac{1}{3} = \frac{4}{12}$, $\frac{1}{12}$ stays as it is. Then $\frac{6}{12} + \frac{4}{12} + \frac{1}{12} = \boxed{\frac{11}{12}}$